



**Generation Interconnection
System Impact Study Report
for
Queue Projects AE1-190 / AE1-191
HARMONY VILLAGE-SHACKLEFORD 115 KV
60 MW Capacity / 100 MW Energy**

Revision 1: March, 2022

Revision 0: August, 2019

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1 Introduction

This System Impact Study (SIS) has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the Feasibility Study Agreement between Carvers Creek LLC, the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is Virginia Electric and Power Company (VEPCO).

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

3 Revision 1 Summary – March 2022

This revision is being issued to incorporate the results of a re-tool and stability analysis.

4 General

The IC has proposed a solar generating facility located in Gloucester County, Virginia. The installed facilities will have a capability of 100 MW with 60 MW of this output being recognized by PJM as Capacity. The

proposed in-service date for the AE1-190/191 project is 12/21/2020. This study does not imply an ITO commitment to either in-service date.

Queue Number	AE1-191
Project Name	HARMONY VILLAGE-SHACKLEFORD 115 KV
Interconnection Customer	Carvers Creek LLC
State	Virginia
County	Gloucester
Transmission Owner	Dominion
MFO	100 (AE1-190: 20, AE1-191:80)
MWE	100 (AE1-190: 20, AE1-191:80)
MWC	60 (AE1-190: 12, AE1-191:48)
Fuel	Solar
Basecase Study Year	2022

4.1 Point of Interconnection

AE1-190/191 will interconnect with the Dominion transmission system as a tap of the Harmony Village to Shackleford 115 kV line. AE1-190/191 will interconnect at the same POI as AD1-041.

4.2 Cost Summary

The AE1-191 project will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$ 0
Direct Connection Network Upgrade	\$ 0
Non Direct Connection Network Upgrades	\$ 0
Total Costs	\$ 0

In addition, the AE1-191 project may be responsible for a contribution to the following costs

Description	Total Cost
System Upgrades	\$ 0

Note: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

5 Transmission Owner Scope of Work

5.1 Attachment Facilities

None

5.2 Direct Connection Cost Estimate

None

6 Non-Direct Connection Cost Estimate

Remote Terminal Work: During the Facilities Study, ITO's System Protection Engineering Department will review transmission line protection as well as anti-islanding required to accommodate the new generation and interconnection substation. System Protection Engineering will determine the minimal acceptable protection requirements to reliably interconnect the proposed generating facility with the transmission system. The review is based on maintaining system reliability by reviewing ITO's protection requirements with the known transmission system configuration which includes generating facilities in the area. This review may determine that transmission line protection and communication upgrades are required at remote substations.

7 Interconnection Customer Requirements

7.1 System Protection

The IC must design its Customer Facilities in accordance with all applicable standards, including the standards in Dominion's "Dominion Energy Electric Transmission Generator Interconnection Requirements" documented in Dominion's Facility Interconnection Requirements "Exhibit C" located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>. Preliminary Protection requirements will be provided as part of the Facilities Study. Detailed Protection Requirements will be provided once the project enters the construction phase.

7.2 Compliance Issues and Interconnection Customer Requirements

The proposed Customer Facilities must be designed in accordance with Dominion's "Dominion's Facility Interconnection Requirements" document located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>. In particular, the IC is responsible for the following:

1. The purchase and installation of a fully rated protection device (circuit breaker, circuit switcher, fuse) to protect the IC's GSU transformer(s).
2. The purchase and installation of the minimum required Dominion generation interconnection relaying and control facilities as described in the System Protection noted above. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.

3. The purchase and installation of supervisory control and data acquisition (“SCADA”) equipment to provide information in a compatible format to the Dominion Transmission System Control Center.
4. Compliance with the Dominion and PJM generator power factor and voltage control requirements.

The GSU(s) associated with the IC queue request shall meet the grounding requirements as noted in Dominion’s “Dominion’s Facility Interconnection Requirements” document located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>.

The IC will also be required to meet all PJM, SERC, and NERC reliability criteria and operating procedures for standards compliance. For example, the IC will need to properly locate and report the over and under voltage and over and under frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and SERC audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the Dominion system.

7.3 Power Factor Requirements

The IC shall design its non-synchronous Customer Facility with the ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.95 lagging (supplying VARs) measured at the high-side of the facility substation transformer(s) connected to the Dominion transmission system.

8 Revenue Metering and SCADA Requirements

8.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

8.1.1 Meteorological Data Reporting Requirement

The solar generation facility shall provide the Transmission Provider with site-specific meteorological data including:

- Temperature (degrees Fahrenheit)
- Atmospheric pressure (hectopascals)
- Irradiance
- Forced outage data

8.2 Dominion Requirements

See Section 3.4.6 “Metering and Telecommunications” of Dominion’s “Dominion’s Facility Interconnection Requirements” document located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>.

9 Network Impacts

The Queue Project AE1-190 / AE1-191 was evaluated as a 100.0 MW (Capacity 60.0 MW) injection via a tap on the Harmony Village to Shackleford 115 kV Line in the Dominion area. Project AE1-191 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AE1-191 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Load Flow

10 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

11 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

12 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

13 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

Contingency			Affected Area	Facility Description	Bus			Power Flow	Loading %		Rating		MW Contribution	Flowgate Appendix
#	Type	Name			From	To	ckt		Initial	Final	Type	MVA		
1	N-1	DVP P1-2: LN 85-A	DVP - DVP	3WEST PT-3LANEXA 115 kV line	314188	314387	1	AC	59.81	103.7	ER	225	99.99	
2	N-1	DVP P1-2: LN 576	DVP - DVP	8ELMONT-8LADYSMITH 500 kV line	314908	314911	1	AC	135.3	136.4	ER	2442	31.37	

14 System Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)

None

15 Flow Gate Details

The following appendices contain additional information about each flowgate presented in the body of the report. For each appendix, a description of the flowgate and its contingency was included for convenience. However, the intent of the appendix section is to provide more information on which projects/generators have contributions to the flowgate in question. Although this information is not used "as is" for cost allocation purposes, it can be used to gage other generators impact. It should be noted the generator contributions presented in the appendices sections are full contributions, whereas in the body of the report, those contributions take into consideration the commercial probability of each project.

15.1 Contingency Descriptions

Contingency Name	Description
DVP_P1-2: LN 576	CONTINGENCY 'DVP_P1-2: LN 576' OPEN BRANCH FROM BUS 314914 TO BUS 314918 CKT 1 /* 8MDLTHAN 500.00 - 8NO ANNA 500.00 END
DVP_P1-2: LN 85-A	CONTINGENCY 'DVP_P1-2: LN 85-A' OPEN BRANCH FROM BUS 314174 TO BUS 934140 CKT 1 /* 3HARMONY 115.00 - AD1-041 TAP 115.00 END

Affected Systems

16 Affected Systems

16.1 Duke Energy Progress

None

Short Circuit

17 Short Circuit

The following breakers are overduty

None

Stability

18 Stability

Generator Interconnection Requests AE1-190 and AE1-191 are for a 20 MW and 80 MW Maximum Facility Output (MFO) solar generation plant, respectively. AE1-190 consists of 8 x 2.5387 MW TMEIC-PVH-L2700GR solar PV inverters and AE1-191 consists of 32 x 2.555 MW TMEIC-PVH-L2700GR solar PV inverters. The Point of Interconnection (POI) is a tap on Harmony Village to Shackleford 115 kV circuit in the Dominion Virginia Power (DVP) transmission system, Gloucester County, Virginia.

The power flow scenario for the analysis was based on the RTEP 2022 summer peak case, modified to include applicable queue projects. AE1-190 and AE1-191 has been dispatched online at maximum facility output, with approximately unity power factor at the high side of the station transformer.

AE1-190 and AE1-191 were tested for compliance with NERC, PJM, and other applicable criteria. The study will include 64 contingencies, each with a 20 second simulation time period. Contingencies to be studied include:

- Steady-state operation (30 second simulation)
- Three-phase faults with normal clearing time
- Single-phase bus faults with normal clearing time
- Single-phase faults with stuck breaker
- Single-phase faults placed at 80% of the line with delayed (Zone 2) clearing at remote line end because of primary communications/relaying failure
- Single-phase faults with loss of multiple circuits caused by a common tower contingency

The 64 fault contingencies tested on the 2022 summer peak case met the recovery criteria:

- AE1-190 and AE1-191 rides through the faults except for faults where protective actions trips a generator(s).
- All generators maintain synchronism and any post-contingency oscillations are positively damped with a damping margin of at least 3%.
- All bus voltages recover to 0.7 p.u. within 2.5 seconds and the final voltages are within the steady-state voltage ranges below per DVP's transmission planning criteria.
 - P1 Category Contingencies:
 - 0.93 to 1.05 p.u. for 230, 138, 115, 69 kV facilities
 - 1.01 to 1.08 p.u. for 500 kV facilities
 - P2, P4, P5, and P7 Category Contingencies:
 - 0.90 to 1.05 p.u. for 230, 138, 115, 69 kV facilities
 - 1.00 to 1.08 p.u. for 500 kV facilities
- No transmission element trips, other than those either directly connected or designated to trip as a consequence of the fault.

The AE1-190 queue project did not meet the 0.95 lagging power factor requirement. An additional 1.82 Mvar is required for the queue project to meet the 0.95 lagging power factor requirement. The AE1-190 queue project did meet the 0.95 leading power factor requirement. See Table 1.

Table 1: Power Factor Assessment for the AE1-190 Queue Project

Generator	MFO (MW)	Required Power Factor Range		Maximum Lagging (Mvar)	Minimum Leading (Mvar)
		Lagging	Leading		
AE1-190	20.00	0.95	0.95		
Total Reactive Power Required				6.57	-6.57
Reactive Power from Generator				Qmax	Qmin
				7.35	-7.35
Customer Planned Compensation				0	0
Reactive Power Losses				-2.60	-2.60
Total Available Reactive Power at High Side of Main Transformer				4.76	-9.95
Deficiency in Reactive Power				-1.82	Meet

The AE1-191 queue project did not meet the 0.95 lagging power factor requirement. An additional 11.28 Mvar is required for the queue project to meet the 0.95 lagging power factor requirement. The AE1-191 queue project did meet the 0.95 leading power factor requirement. See Table 2.

Table 2: Power Factor Assessment for the AE1-191 Queue Project

Generator	MFO (MW)	Required Power Factor Range		Maximum Lagging (Mvar)	Minimum Leading (Mvar)
		Lagging	Leading		
AE1-191	80.00	0.95	0.95		
Total Reactive Power Required				26.29	-26.29
Reactive Power from Generator				Qmax	Qmin
				27.90	-27.90
Customer Planned Compensation				0	0
Reactive Power Losses				-12.89	-12.89
Total Available Reactive Power at High Side of Main Transformer				15.01	-40.80
Deficiency in Reactive Power				-11.28	Meet

System Configuration

